EXECUTIVE SUMMARY

Weightlifters have access to countless fitness apps and wearable devices, yet none provide accurate, real-time feedback on lifting form, vitals, and performance in a single, portable system. Poor form remains a leading cause of preventable gym injuries, and most lifters have no affordable way to monitor technique or track physiological data while exercising. The opportunity exists for a device that enhances safety, improves training efficiency, and provides meaningful performance insight without requiring a coach or complex setup. The R.E.P. (Repetition, Efficiency, Precision) Tracker, depicted in Figure 1, addresses this gap by combining motion tracking, vitals monitoring, and live video analysis into a compact, cordless smart-coaching system designed specifically for weightlifting environments.

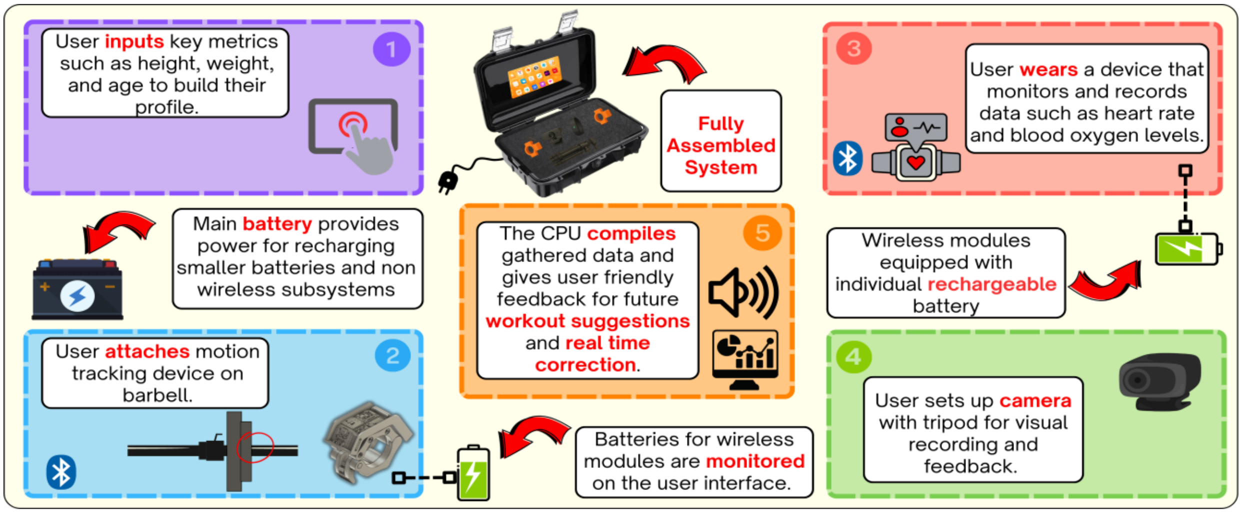

Figure 1. Overview of R.E.P Tracker

The R.E.P Tracker project has a few core system requirements and constraints that are vital to its operation and distribution. The system is required to attach to a standard or Olympic barbell. When counting and recording rep data, there must be a maximum error of 2% for raw data. The battery life of each submodule should be able to run for a minimum of 4 hours under a full load to ensure proper function during extended use. A main constraint for the system is the range of clear Bluetooth communication between subsystems within 30ft. Likewise, it should be as light as possible. Thus, the entire system must have a total weight of 22.05lb (10kg) or less. The system must follow the standards for battery safety (IEC 62368-1) and Bluetooth communication (IEEE 802.15.1) to ensure safety from shocks or data latency.

The overall system provides real-time form analysis and vitals monitoring to enhance weightlifting safety and effectiveness. Each sensor was chosen to meet the constraints and requirements that ultimately were identified as essential to the user's experience with the system. Each subsystem – Motion Tracking (MT), User Interface (UI), Power System (PS) – was chosen to maximize the market niche and utility to the exerciser. The central part chosen for MT is the Adafruit Breakout Board with BNO055, acquired because of its accuracy and automated calibration. The pivotal component chosen for the UI was the Hosyond 1024x600 Touchscreen display. This touchscreen has both compatibility with the central processor used (Raspberry Pi 5) and a clear, interactive display for the user. The PS main component is the the Eryy 12.8V 15Ah LiFePo4 main battery. The main battery was chosen because it can sustain multiple charging cycles for the whole system while simultaneously being able to provide the central processor with sufficient power. For the UI, QT Designer software was chosen for screen development due to its simplicity and compatibility with the project's systems. The overall project is mostly programmed in Python because of its coding library support and compatibility with the project's goals and subsystems. The next step will be implementing 3 more subsystems in the Spring: Performance Monitoring, Camera Analysis, and User Feedback

The R.E.P Tracker can still be expanded in many ways; it can grow in workout variety along with even more unique and personalized feedback. It will also open the path to AI-driven personalization in the field, helping to keep workouts safe along with assisting in rehabilitation. This will be a great help to other existing fields, such as Kinesiology Research, by providing a dependable device that is a reliable driver of statistics.